PHẦN TỬ DSG3

PHÁT TRIỂN PHẦN TỬ TẤM REISSNER\MINDLIN THÀNH PHẦN TỬ FLAT SHELL

Trình bày: Phùng Văn Phúc

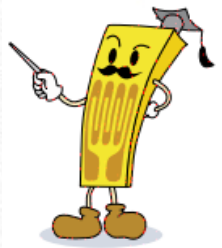
GVHD: TS. Nguyễn Thời Trung

Tp. HCM 11-08-2011

Nội dung trình bày



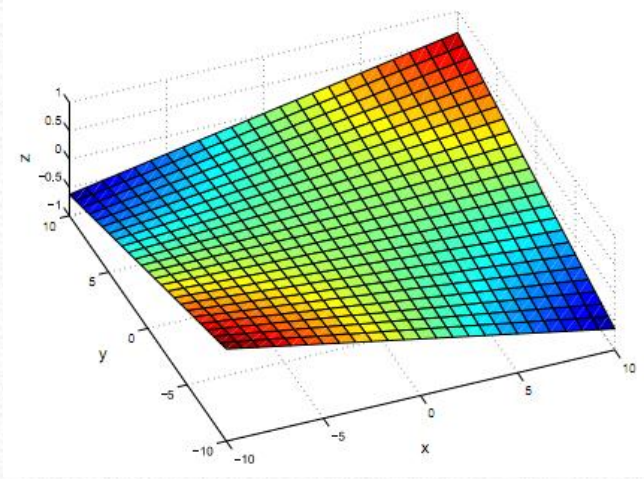
- Phần tử shell
- Thành lập Bm (biến dạng màng)
- Thành lập Bb (biến dạng uốn)
- Thành lập Bs (biến dạng cắt)
- Chuyển đổi tọa độ

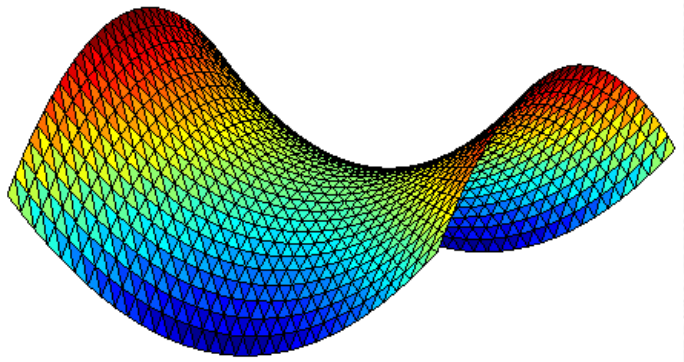


Tổng quan shell

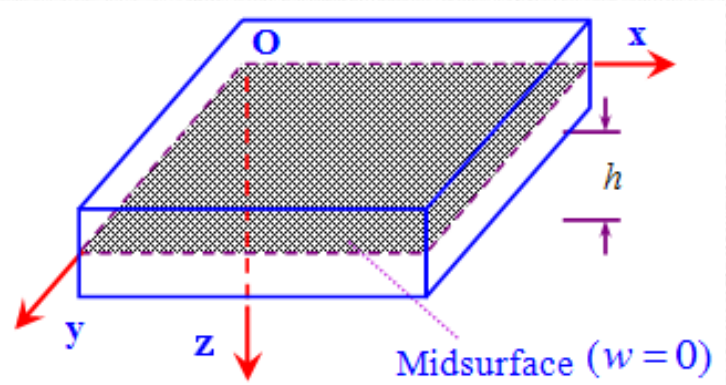


➢ Shell





➢ Plate



- Shell sử dụng cho vật thể có độ cong về hình học (hình học phức tạp)
- Classical shell → phức tạp
- Plat shell



Plat shell



➢ Phát triển Plate lên Flat shell

Plate (w, β_y, β_x) + Membrane (u, v) + Coordinate Transformation → Flat shell (u, v, w, β_y, β_x, β_z)

- β_x là góc quay xung quanh trục y
- β_y là góc quay xung quanh trục x
- Membrane biến dạng màng (ứng suất phẳng hoặc biến dạng phẳng)
- Coordinate Transformation quan hệ giữa tọa độ địa phương (tính K phần tử) và hệ tọa độ toàn cục (lắp ghép K)



Plat shell



- Membrame element
  - Biến dạng trong mặt phẳng, bài toán ứng suất phẳng hoặc bài toán biến dạng phẳng

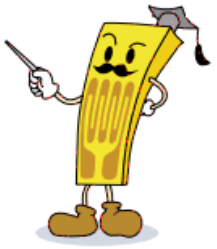
$$\varepsilon_m = \left[\frac{\partial u}{\partial x} \quad \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right]^T$$

- Bending element
  - Biến dạng uốn do góc xoay gây ra (Kirchoff)

$$\varepsilon_b = z \left[\frac{\partial \beta_x}{\partial x} \quad \frac{\partial \beta_y}{\partial y} \quad \frac{\partial \beta_x}{\partial y} + \frac{\partial \beta_y}{\partial x} \right]^T = z\kappa^T$$

- Shear element
  - Biến dạng cắt

$$\gamma = \begin{bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{bmatrix} = \begin{bmatrix} w'_x + \beta_x \\ w'_y + \beta_y \end{bmatrix}$$

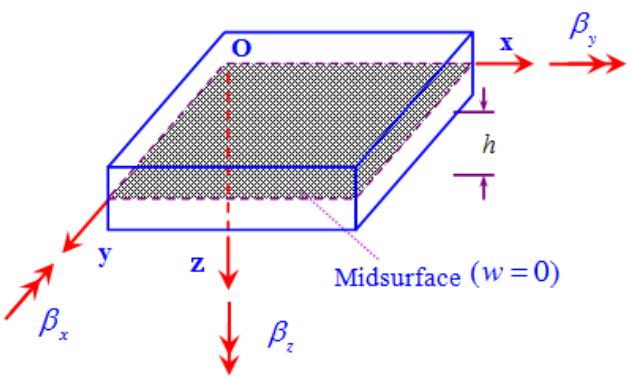


Plat shell



➢ Shell: mỗi node có 6 bậc tự do

$$\mathbf{U}_{\beta} = \begin{bmatrix} u \\ v \\ w \\ \beta_x \\ \beta_y \\ \beta_z \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 & 0 & 0 & 0 \\ 0 & N_i & 0 & 0 & 0 & 0 \\ 0 & 0 & N_i & 0 & 0 & 0 \\ 0 & 0 & 0 & N_i & 0 & 0 \\ 0 & 0 & 0 & 0 & N_i & 0 \\ 0 & 0 & 0 & 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ w_i \\ \beta_{xi} \\ \beta_{yi} \\ \beta_{zi} \end{bmatrix}$$



○ Phát triển tấm DSG3 thành flat shell DSG3 (phần tử flat shell ba node)



Plat shell

➢ Membrame element

$$\varepsilon_m = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \underbrace{\begin{bmatrix} \dfrac{\partial N_i}{\partial x} & 0 & 0 & 0 & 0 & 0 \\ 0 & \dfrac{\partial N_i}{\partial y} & 0 & 0 & 0 & 0 \\ \dfrac{\partial N_i}{\partial y} & \dfrac{\partial N_i}{\partial x} & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{B}_m} \begin{bmatrix} u_i \\ v_i \\ w_i \\ \beta_{xi} \\ \beta_{yi} \\ \beta_{zi} \end{bmatrix}$$

➢ Theo lý thuyết tấm DSG3

$$\frac{\partial N_1}{\partial x} = \frac{b-c}{\det \mathbf{J}} \qquad \frac{\partial N_1}{\partial y} = \frac{d-a}{\det \mathbf{J}}$$

$$\frac{\partial N_2}{\partial x} = \frac{c}{\det \mathbf{J}} \qquad \frac{\partial N_2}{\partial y} = \frac{-d}{\det \mathbf{J}}$$

$$\frac{\partial N_3}{\partial x} = \frac{-b}{\det \mathbf{J}} \qquad \frac{\partial N_3}{\partial y} = \frac{a}{\det \mathbf{J}}$$

$$\mathbf{B}_m = \frac{1}{\det \mathbf{J}} \begin{bmatrix} b-c & 0 & 0 & 0 & 0 & 0 & c & 0 & 0 & 0 & 0 & 0 & -b & 0 & 0 & 0 & 0 & 0 \\ 0 & d-a & 0 & 0 & 0 & 0 & 0 & -d & 0 & 0 & 0 & 0 & 0 & a & 0 & 0 & 0 & 0 \\ d-a & b-c & 0 & 0 & 0 & 0 & -d & c & 0 & 0 & 0 & 0 & a & -b & 0 & 0 & 0 & 0 \end{bmatrix}$$

Ứng suất phẳng

$$\mathbf{D}_m = \frac{Et}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}$$

$$\mathbf{K}_m = \mathbf{B}_m^T \mathbf{D}_m \mathbf{B}_m$$

$$\mathbf{D}_m = \frac{Et}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & 0 & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix}$$

Biến dạng phẳng

Plat shell

➢ Bending element

$$\varepsilon_b = \begin{bmatrix} \frac{\partial \beta_x}{\partial x} \\ \frac{\partial \beta_y}{\partial y} \\ \frac{\partial \beta_y}{\partial y} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & \frac{\partial N_i}{\partial x} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\partial N_i}{\partial y} & 0 \\ 0 & 0 & 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} & 0 \end{bmatrix}}_{\mathbf{B}_b} \begin{bmatrix} u_i \\ v_i \\ w_i \\ \beta_{xi} \\ \beta_{yi} \\ \beta_{zi} \end{bmatrix}$$

➢ Theo lý thuyết tấm DSG3

$$\frac{\partial N_1}{\partial x} = \frac{b-c}{\det \mathbf{J}} \qquad \frac{\partial N_1}{\partial y} = \frac{d-a}{\det \mathbf{J}}$$

$$\frac{\partial N_2}{\partial x} = \frac{c}{\det \mathbf{J}} \qquad \frac{\partial N_2}{\partial y} = \frac{-d}{\det \mathbf{J}}$$

$$\frac{\partial N_3}{\partial x} = \frac{-b}{\det \mathbf{J}} \qquad \frac{\partial N_3}{\partial y} = \frac{a}{\det \mathbf{J}}$$

$$\mathbf{B}_b = \frac{1}{\det \mathbf{J}} \begin{bmatrix} 0 & 0 & 0 & b-c & 0 & 0 & 0 & 0 & c & 0 & 0 & 0 & 0 & -b & 0 & 0 \\ 0 & 0 & 0 & 0 & d-a & 0 & 0 & 0 & 0 & -d & 0 & 0 & 0 & 0 & a & 0 \\ 0 & 0 & 0 & d-a & b-c & 0 & 0 & 0 & -d & c & 0 & 0 & 0 & a & -b & 0 \end{bmatrix}$$

$$\mathbf{K}_b = \mathbf{B}_b^T \mathbf{D}_b \mathbf{B}_b$$

Plat shell



- Theo lý thuyết tấm DSG3

$$\mathbf{B}^s = \frac{1}{\det \mathbf{J}} \begin{bmatrix} b-c & \frac{\det \mathbf{J}}{2} & 0 & c & \frac{ac}{2} & \frac{bc}{2} & -b & -\frac{bd}{2} & -\frac{bc}{2} \\ d-a & 0 & \frac{\det \mathbf{J}}{2} & -d & -\frac{ad}{2} & -\frac{bd}{2} & a & \frac{ad}{2} & \frac{ac}{2} \end{bmatrix}$$

- Shear element

$$\mathbf{B}^s = \frac{1}{\det \mathbf{J}} \begin{bmatrix} 0 & 0 & b-c & \frac{\det \mathbf{J}}{2} & 0 & 0 & 0 & 0 & c & \frac{ac}{2} & \frac{bc}{2} & 0 & 0 & 0 & -b & -\frac{bd}{2} & -\frac{bc}{2} & 0 \\ 0 & 0 & d-a & 0 & \frac{\det \mathbf{J}}{2} & 0 & 0 & 0 & -d & -\frac{ad}{2} & -\frac{bd}{2} & 0 & 0 & 0 & a & \frac{ad}{2} & \frac{ac}{2} & 0 \end{bmatrix}$$

$$\mathbf{K}_s = \mathbf{B}_s^T \mathbf{D}_s \mathbf{B}_s$$



Khử bậc tự do xoay quanh trục z (drilling)



➢ Ma trận độ cứng trong plat shell

$$\mathbf{K}=\mathbf{B}_m^T\mathbf{D}_m\mathbf{B}_m+\mathbf{B}_b^T\mathbf{D}_b\mathbf{B}_b+\mathbf{B}_s^T\mathbf{D}_s\mathbf{B}_s=\mathbf{K}_m+\mathbf{K}_b+\mathbf{K}_s$$

○ Ma trận K bị suy biến tại vị trí bậc tự do xoay quanh trục z (drilling rotation)

✓ Triangular plat shell elements (15 bậc tự do)

$$\mathbf{K}=\mathbf{K}_m+\mathbf{K}_b+\mathbf{K}_s+\mathbf{K}_\beta$$

$$k_\beta=\alpha EA\begin{bmatrix}1 & -0.5 & -0.5\\ -0.5 & 1 & -0.5\\ -0.5 & -0.5 & 1\end{bmatrix}$$

$\alpha = t\times 10^{-4}$

E *elastic module*

A *area of the element*

$$\mathbf{K}_\beta(in,in)_{18\times 18}=\mathbf{K}_\beta(in,in)+k_\beta \quad ; \quad in=(6,12,18)$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

○ Phần drilling được cộng vào biến dạng màng (membrame)

$$u = \sum_{i=1}^{3} N_i u_i + p_1\beta_{z1} + p_2\beta_{z2} + p_3\beta_{z3} \quad ; \quad v = \sum_{i=1}^{3} N_i v_i + q_1\beta_{z1} + q_2\beta_{z2} + q_3\beta_{z3}$$

$$p_1 = (b_3N_2 - b_2N_3)\,N_1/2 \qquad p_2 = (b_1N_3 - b_3N_1)\,N_2/2 \qquad p_3 = (b_2N_1 - b_1N_2)\,N_3/2$$

$$q_1 = (c_3N_2 - c_2N_3)\,N_1/2 \qquad q_2 = (c_1N_3 - c_3N_1)\,N_2/2 \qquad q_3 = (c_2N_1 - c_1N_2)\,N_3/2$$

$$b_1 = y_2 - y_3 \quad b_2 = y_3 - y_1 \quad b_3 = y_1 - y_2$$

$$c_1 = x_3 - x_2 \quad c_2 = x_1 - x_3 \quad c_3 = x_2 - x_1$$

○ Biến dạng màng (membrame)

$$\varepsilon_m = \left[\frac{\partial u}{\partial x} \quad \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right]$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

$$\frac{\partial p_1}{\partial x}=\frac{\partial}{\partial x}\left[(b_3N_2-b_2N_3)\frac{N_1}{2}\right]=\left(b_3\frac{\partial N_2}{\partial x}-b_2\frac{\partial N_3}{\partial x}\right)\frac{N_1}{2}+(b_3N_2-b_2N_3)\frac{\partial N_1}{2\partial x}$$

$$=\left(b_3\frac{\mathrm{c}}{\det\mathbf{J}}+b_2\frac{\mathrm{b}}{\det\mathbf{J}}\right)\frac{N_1}{2}+(b_3N_2-b_2N_3)\frac{\mathrm{b}-\mathrm{c}}{2\det\mathbf{J}}$$

$$\frac{\partial p_2}{\partial x}=\frac{\partial}{\partial x}\left[(b_1N_3-b_3N_1)\frac{N_2}{2}\right]=\left(b_1\frac{\partial N_3}{\partial x}-b_3\frac{\partial N_1}{\partial x}\right)\frac{N_2}{2}+(b_1N_3-b_3N_1)\frac{\partial N_2}{2\partial x}$$

$$=\left(b_1\frac{-\mathrm{b}}{\det\mathbf{J}}-b_3\frac{\mathrm{b}-\mathrm{c}}{\det\mathbf{J}}\right)\frac{N_2}{2}+(b_1N_3-b_3N_1)\frac{\mathrm{c}}{2\det\mathbf{J}}$$

$$\frac{\partial p_3}{\partial x}=\frac{\partial}{\partial x}\left[(b_2N_1-b_1N_2)\frac{N_3}{2}\right]=\left(b_2\frac{\partial N_1}{\partial x}-b_1\frac{\partial N_2}{\partial x}\right)\frac{N_3}{2}+(b_2N_1-b_1N_2)\frac{\partial N_3}{2\partial x}$$

$$=\left(b_2\frac{\mathrm{b}-\mathrm{c}}{\det\mathbf{J}}-b_1\frac{\mathrm{c}}{\det\mathbf{J}}\right)\frac{N_3}{2}+(b_2N_1-b_1N_2)\frac{-\mathrm{b}}{2\det\mathbf{J}}$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

$$\frac{\partial p_1}{\partial y}=\frac{\partial}{\partial y}\left[(b_3N_2-b_2N_3)\frac{N_1}{2}\right]=\left(b_3\frac{\partial N_2}{\partial y}-b_2\frac{\partial N_3}{\partial y}\right)\frac{N_1}{2}+(b_3N_2-b_2N_3)\frac{\partial N_1}{2\partial y}$$

$$=\left(b_3\frac{-\mathrm{d}}{\det\mathbf{J}}-b_2\frac{\mathrm{a}}{\det\mathbf{J}}\right)\frac{N_1}{2}+(b_3N_2-b_2N_3)\frac{\mathrm{d}-\mathrm{a}}{2\det\mathbf{J}}$$

$$\frac{\partial p_2}{\partial y}=\frac{\partial}{\partial y}\left[(b_1N_3-b_3N_1)\frac{N_2}{2}\right]=\left(b_1\frac{\partial N_3}{\partial y}-b_3\frac{\partial N_1}{\partial y}\right)\frac{N_2}{2}+(b_1N_3-b_3N_1)\frac{\partial N_2}{2\partial y}$$

$$=\left(b_1\frac{\mathrm{a}}{\det\mathbf{J}}-b_3\frac{\mathrm{d}-\mathrm{a}}{\det\mathbf{J}}\right)\frac{N_2}{2}+(b_1N_3-b_3N_1)\frac{-\mathrm{d}}{2\det\mathbf{J}}$$

$$\frac{\partial p_3}{\partial y}=\frac{\partial}{\partial y}\left[(b_2N_1-b_1N_2)\frac{N_3}{2}\right]=\left(b_2\frac{\partial N_1}{\partial y}-b_1\frac{\partial N_2}{\partial y}\right)\frac{N_3}{2}+(b_2N_1-b_1N_2)\frac{\partial N_3}{2\partial y}$$

$$=\left(b_2\frac{\mathrm{d}-\mathrm{a}}{\det\mathbf{J}}+b_1\frac{\mathrm{d}}{\det\mathbf{J}}\right)\frac{N_3}{2}+(b_2N_1-b_1N_2)\frac{\mathrm{a}}{2\det\mathbf{J}}$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

$$\frac{\partial q_1}{\partial x}=\frac{\partial}{\partial x}\left[(c_3N_2-c_2N_3)\frac{N_1}{2}\right]=\left(c_3\frac{\partial N_2}{\partial x}-c_2\frac{\partial N_3}{\partial x}\right)\frac{N_1}{2}+(c_3N_2-c_2N_3)\frac{\partial N_1}{2\partial x}$$

$$=\left(c_3\frac{\mathrm{c}}{\det\mathbf{J}}+c_2\frac{\mathrm{b}}{\det\mathbf{J}}\right)\frac{N_1}{2}+(c_3N_2-c_2N_3)\frac{\mathrm{b}-\mathrm{c}}{2\det\mathbf{J}}$$

$$\frac{\partial q_2}{\partial x}=\frac{\partial}{\partial x}\left[(c_1N_3-c_3N_1)\frac{N_2}{2}\right]=\left(c_1\frac{\partial N_3}{\partial x}-c_3\frac{\partial N_1}{\partial x}\right)\frac{N_2}{2}+(c_1N_3-c_3N_1)\frac{\partial N_2}{2\partial x}$$

$$=\left(c_1\frac{-\mathrm{b}}{\det\mathbf{J}}-c_3\frac{\mathrm{b}-\mathrm{c}}{\det\mathbf{J}}\right)\frac{N_2}{2}+(c_1N_3-c_3N_1)\frac{\mathrm{c}}{2\det\mathbf{J}}$$

$$\frac{\partial q_3}{\partial x}=\frac{\partial}{\partial x}\left[(c_2N_1-c_1N_2)\frac{N_3}{2}\right]=\left(c_2\frac{\partial N_1}{\partial x}-c_1\frac{\partial N_2}{\partial x}\right)\frac{N_3}{2}+(c_2N_1-c_1N_2)\frac{\partial N_3}{2\partial x}$$

$$=\left(c_2\frac{\mathrm{b}-\mathrm{c}}{\det\mathbf{J}}-c_1\frac{\mathrm{c}}{\det\mathbf{J}}\right)\frac{N_3}{2}+(c_2N_1-c_1N_2)\frac{-\mathrm{b}}{2\det\mathbf{J}}$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

$$\frac{\partial q_1}{\partial y}=\frac{\partial}{\partial y}\left[(c_3N_2-c_2N_3)\frac{N_1}{2}\right]=\left(c_3\frac{\partial N_2}{\partial y}-c_2\frac{\partial N_3}{\partial y}\right)\frac{N_1}{2}+(c_3N_2-c_2N_3)\frac{\partial N_1}{2\partial y}$$

$$=\left(c_3\frac{-\mathrm{d}}{\det\mathbf{J}}-c_2\frac{\mathrm{a}}{\det\mathbf{J}}\right)\frac{N_1}{2}+(c_3N_2-c_2N_3)\frac{\mathrm{d}-\mathrm{a}}{2\det\mathbf{J}}$$

$$\frac{\partial q_2}{\partial y}=\frac{\partial}{\partial y}\left[(c_1N_3-c_3N_1)\frac{N_2}{2}\right]=\left(c_1\frac{\partial N_3}{\partial y}-c_3\frac{\partial N_1}{\partial y}\right)\frac{N_2}{2}+(c_1N_3-c_3N_1)\frac{\partial N_2}{2\partial y}$$

$$=\left(c_1\frac{\mathrm{a}}{\det\mathbf{J}}-c_3\frac{\mathrm{d}-\mathrm{a}}{\det\mathbf{J}}\right)\frac{N_2}{2}+(c_1N_3-c_3N_1)\frac{-\mathrm{d}}{2\det\mathbf{J}}$$

$$\frac{\partial q_3}{\partial y}=\frac{\partial}{\partial y}\left[(c_2N_1-c_1N_2)\frac{N_3}{2}\right]=\left(c_2\frac{\partial N_1}{\partial y}-c_1\frac{\partial N_2}{\partial y}\right)\frac{N_3}{2}+(c_2N_1-c_1N_2)\frac{\partial N_3}{2\partial y}$$

$$=\left(c_2\frac{\mathrm{d}-\mathrm{a}}{\det\mathbf{J}}+c_1\frac{\mathrm{d}}{\det\mathbf{J}}\right)\frac{N_3}{2}+(c_2N_1-c_1N_2)\frac{\mathrm{a}}{2\det\mathbf{J}}$$





Khử bậc tự do xoay quanh trục z (drilling)

✓ Triangular plat shell elements with fictitious drilling ratation (18P *dof)*

$$\varepsilon_m = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 & 0 & 0 & 0 & \frac{\partial p_i}{\partial x} \\ 0 & \frac{\partial N_i}{\partial y} & 0 & 0 & 0 & \frac{\partial q_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} & 0 & 0 & 0 & \frac{\partial p_i}{\partial y} + \frac{\partial q_i}{\partial x} \end{bmatrix}}_{\mathbf{B}_m} \begin{bmatrix} u_i \\ v_i \\ w_i \\ \beta_{xi} \\ \beta_{yi} \\ \beta_{zi} \end{bmatrix}$$

$$\mathbf{B}_m^{18P} = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & 0 & 0 & \frac{\partial p_1}{\partial x} & \frac{\partial N_2}{\partial x} & 0 & 0 & 0 & 0 & \frac{\partial p_2}{\partial x} & \frac{\partial N_3}{\partial x} & 0 & 0 & 0 & 0 & \frac{\partial p_3}{\partial x} \\ 0 & \frac{\partial N_1}{\partial y} & 0 & 0 & 0 & \frac{\partial q_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & 0 & 0 & \frac{\partial q_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & 0 & 0 & \frac{\partial q_3}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & 0 & 0 & \frac{\partial p_1}{\partial y} + \frac{\partial q_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & 0 & 0 & 0 & \frac{\partial p_2}{\partial y} + \frac{\partial q_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & 0 & 0 & 0 & \frac{\partial p_3}{\partial y} + \frac{\partial q_3}{\partial x} \end{bmatrix}$$



Khử bậc tự do xoay quanh trục z (drilling)



✓ Triangular plat shell elements with true drilling ratation (18T *dof)*

○ Xấp xỉ bậc tự do màng(membrame)

$$\mathbf{u}=\begin{pmatrix}u\\v\end{pmatrix}=\underbrace{\begin{bmatrix}N_i & 0 & 0 & 0 & 0 & 0\\ 0 & N_i & 0 & 0 & 0 & 0\end{bmatrix}}_{\mathbf{F}}\underbrace{\begin{bmatrix}u_i\\v_i\\w_i\\\beta_{xi}\\\beta_{yi}\\\beta_{zi}\end{bmatrix}}_{\mathbf{U}_\beta}=\underset{2\times 18}{\mathbf{F}}\,\mathbf{U}_\beta$$

○ Xấp xỉ bậc tự do drilling

$$\beta_z=\sum_{i=1}^{3}N_i\beta_{zi}=\underbrace{\begin{bmatrix}0 & 0 & 0 & 0 & 0 & N_i\end{bmatrix}}_{\mathbf{F}_\beta}\underbrace{\begin{bmatrix}u_i\\v_i\\w_i\\\beta_{xi}\\\beta_{yi}\\\beta_{zi}\end{bmatrix}}_{\mathbf{U}_\beta}=\underset{1\times 18}{\mathbf{F}_\beta}\,\mathbf{U}_\beta$$



Khử bậc tự do xoay quanh trục z (drilling)



- ✓ Triangular plat shell elements with true drilling ratation (18T *dof)*
  - ○ Năng lượng biến dạng

$$\Pi = \Pi_m + \Pi_b + \Pi_s + \int_\Omega \alpha \frac{Et}{2(1-\nu^2)} \left(\beta_z - \bar{\mathbf{L}}u\right)^T \left(\beta_z - \bar{\mathbf{L}}u\right) \mathrm{d}\Omega$$

$$\bar{\mathbf{L}} = \frac{1}{2}\begin{bmatrix} -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \qquad \alpha = 10^{-4} \quad \textit{penalty factor}$$

  - ○ Ma trận độ cứng hệ

$$\mathbf{K} = \mathbf{K}_m + \mathbf{K}_b + \mathbf{K}_s + \mathbf{K}_\beta$$

$$\mathbf{K}_\beta = \alpha \frac{Et}{1-\nu^2} \int_\Omega \mathrm{B}_\beta^T \mathrm{B}_\beta \mathrm{d}\Omega \qquad \underset{1\times 18}{\mathrm{B}_\beta} = \underset{1\times 18}{\mathbf{F}_\beta} - \underset{1\times 2}{\bar{\mathbf{L}}} \; \underset{2\times 18}{\mathbf{F}}$$



Khử bậc tự do xoay quanh trục z (drilling)



- ✓ Triangular plat shell elements with true drilling ratation (18T *dof)*



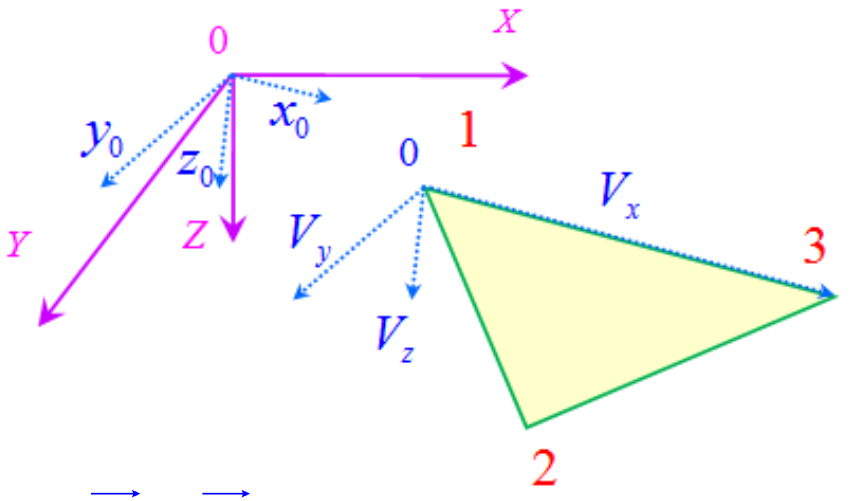
Plat shell

- Coordinate transformation
  - Phần tử tam giác

✓ Hệ tọa độ global XYZ

✓ Hệ tọa độ local $x_0y_0z_0$

✓ *Xác định hệ local*



- Chọn 1 cạnh là trục theo phương x $\overrightarrow{V_x} = \overrightarrow{13}$

- Xác định trục z local $\overrightarrow{V_z} = \overrightarrow{V_x} \times \overrightarrow{12}$

- Xác định trục y local $\overrightarrow{V_y} = \overrightarrow{V_z} \times \overrightarrow{V_x}$

✓ *Xác định các cosin*

$$\lambda_x = \frac{\overrightarrow{V_x}}{|V_x|} \qquad \lambda_y = \frac{\overrightarrow{V_y}}{|V_y|} \qquad \lambda_z = \frac{\overrightarrow{V_z}}{|V_z|}$$

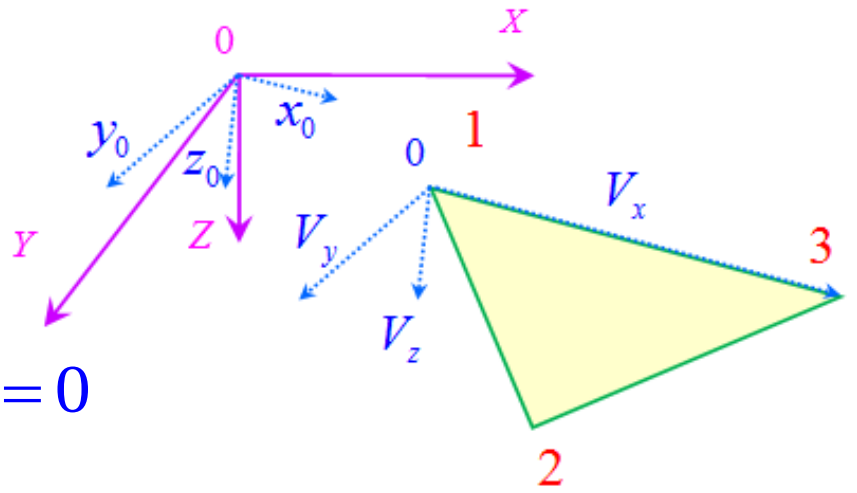
$$[\lambda]_{3\times3} = \left[(\lambda_x)_{3\times1} \quad (\lambda_y)_{3\times1} \quad (\lambda_z)_{3\times1} \right]$$

✓ *Quan hệ local và global*

$$[x_0 y_0 z_0]_{3\times1} = [\lambda]^T_{3\times3} [XYZ]_{3\times1}$$

Plat shell

- Coordinate transformation
  - *Tọa độ các nút trong local*
    - Nút 1



$$xprime(1) = 0 \qquad yprime(1) = 0$$

    - Nút 3

$$xprime(3) = |13| \qquad yprime(3) = 0$$

    - Nút 2

$$xprime(2) = |12|\cos\alpha$$

$$yprime(2) = |12|\sin\alpha$$

$$\cos\alpha = \frac{|12|^2 + |13|^2 - |23|^2}{2|12||13|}$$



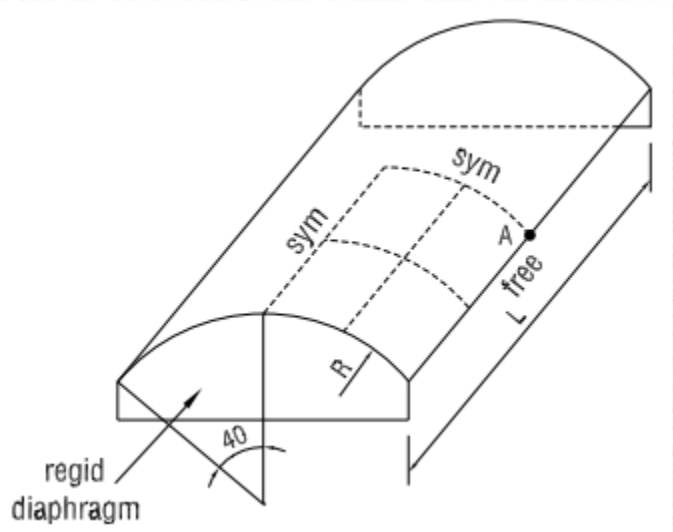
PHẦN TỬ DSG3

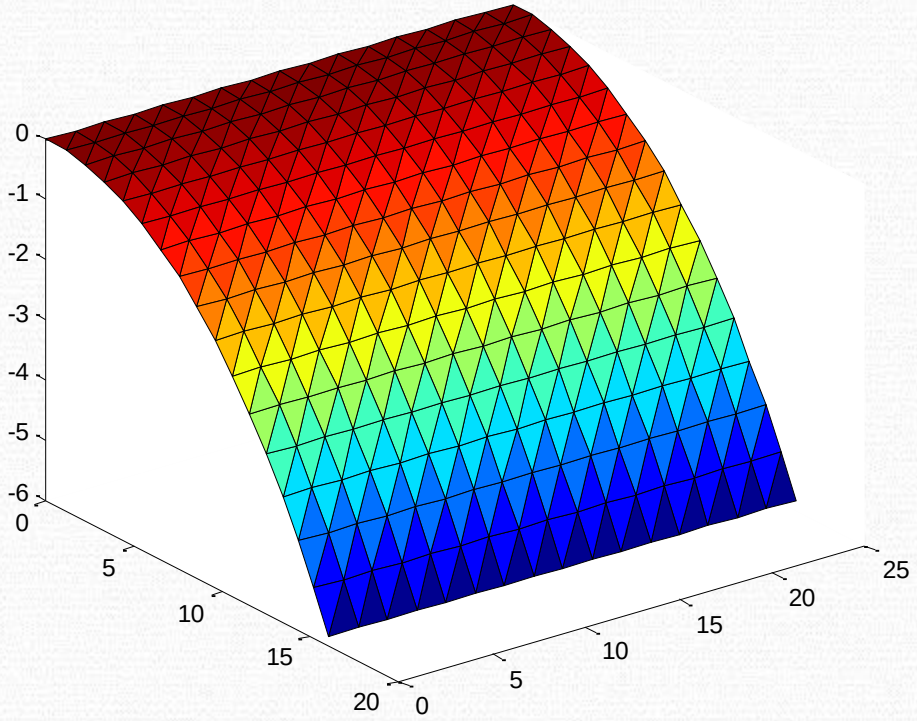
MỘT SỐ KẾT QUẢ FLAT SHELL DÙNG PHƯƠNG PHÁP CS-DSG3



Plat shell

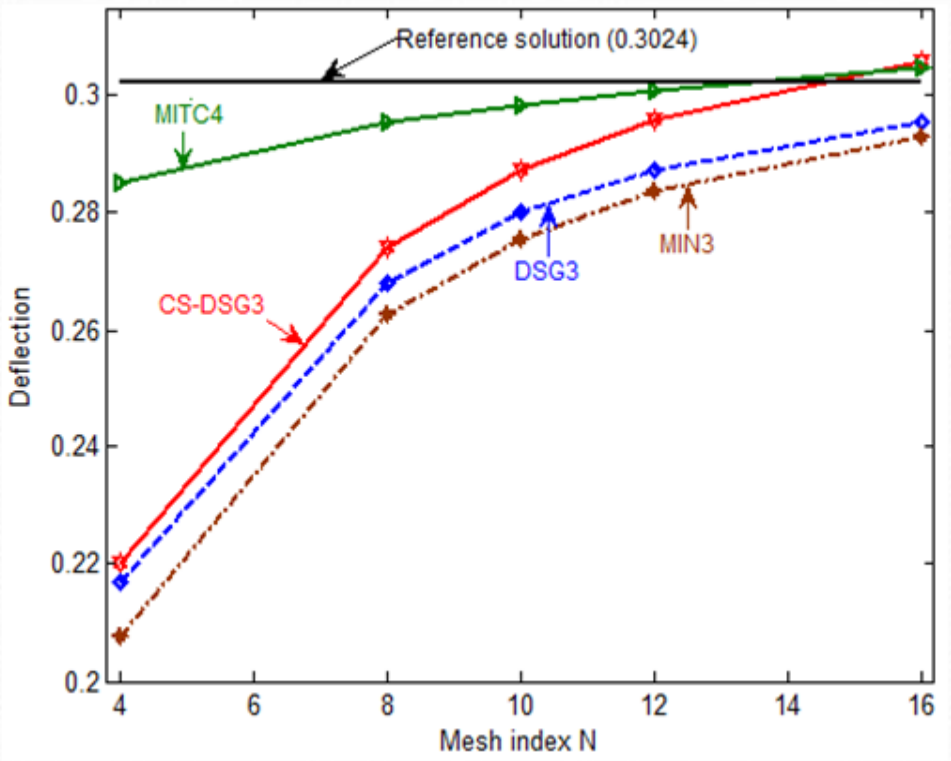
Scordedis Lo roof

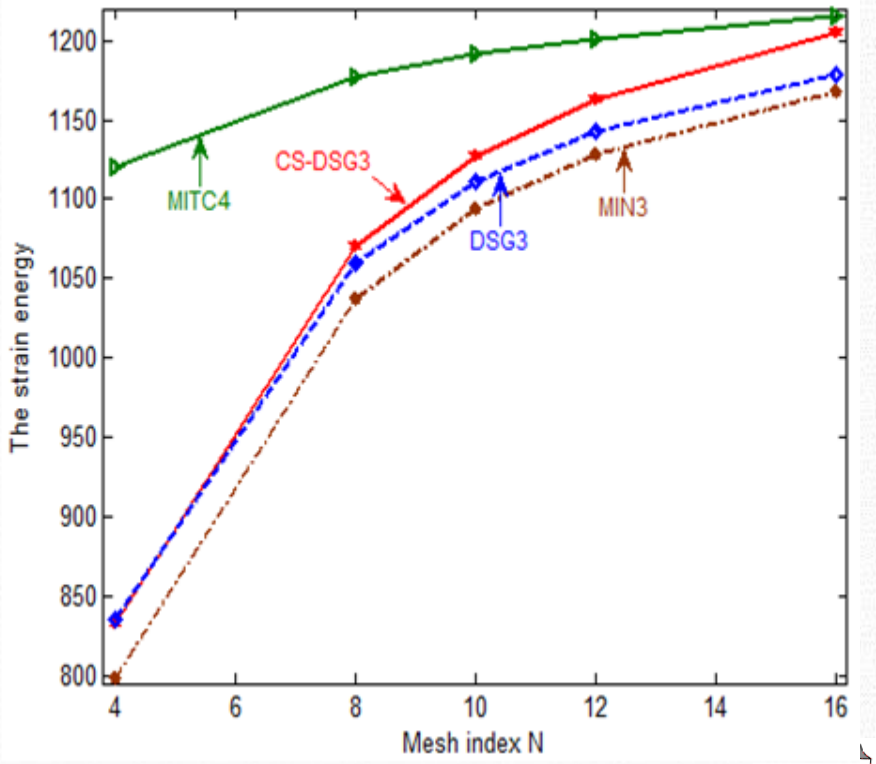




Plat shell

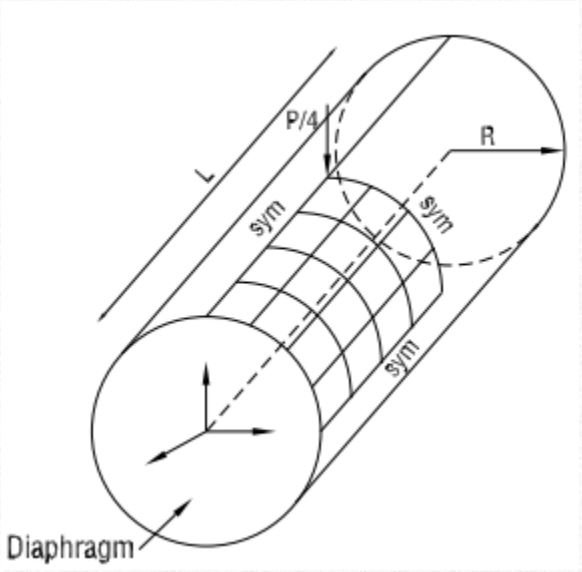
Scordedis Lo roof

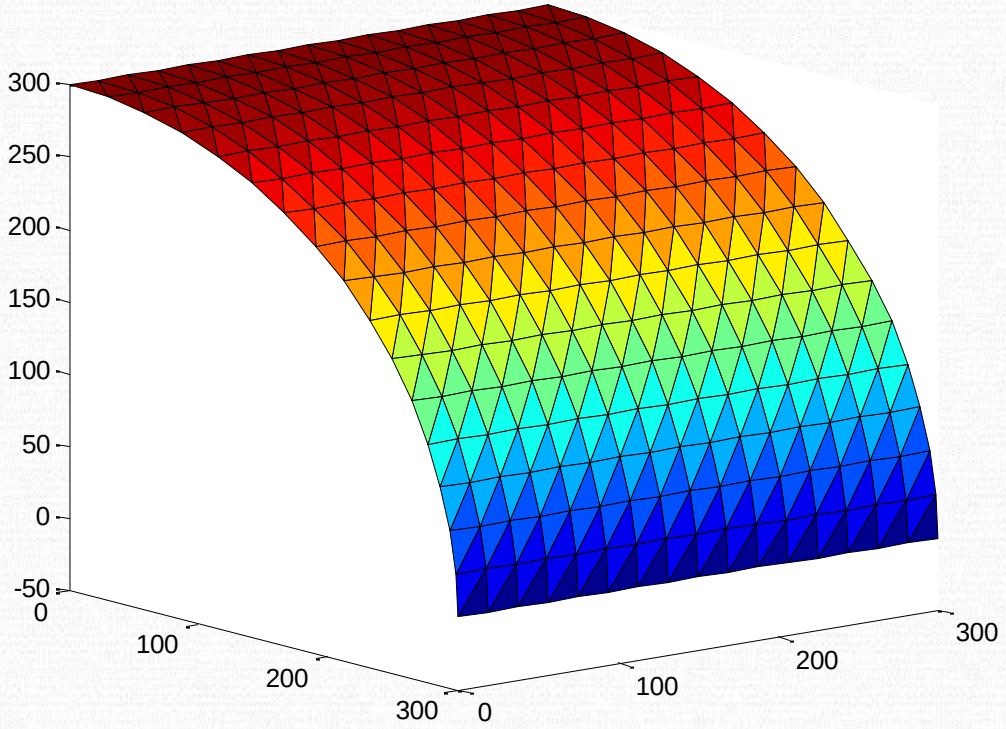




Plat shell

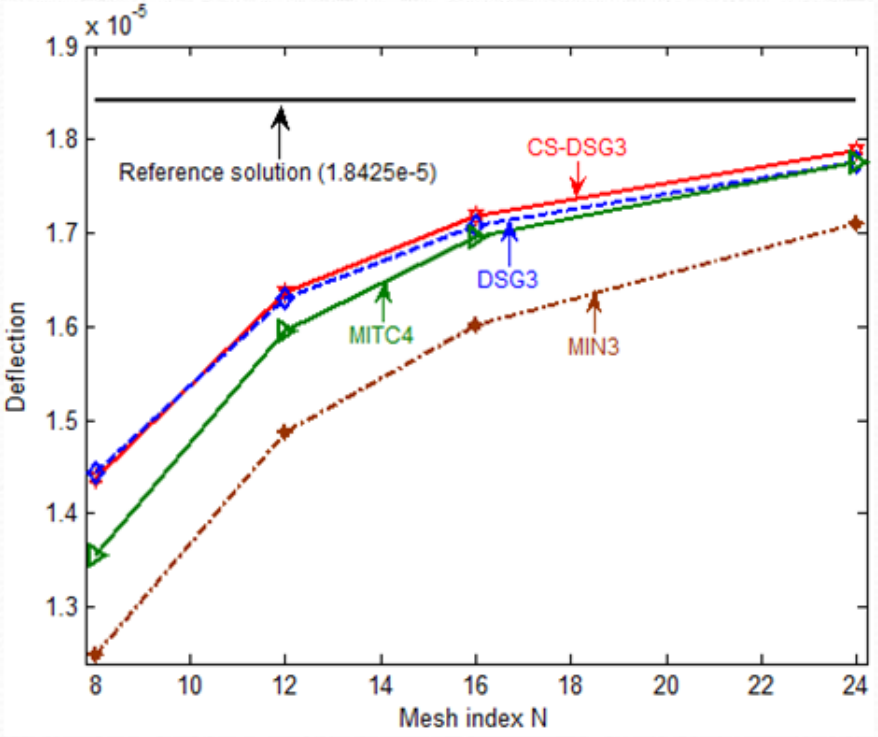
Pinched cylinder

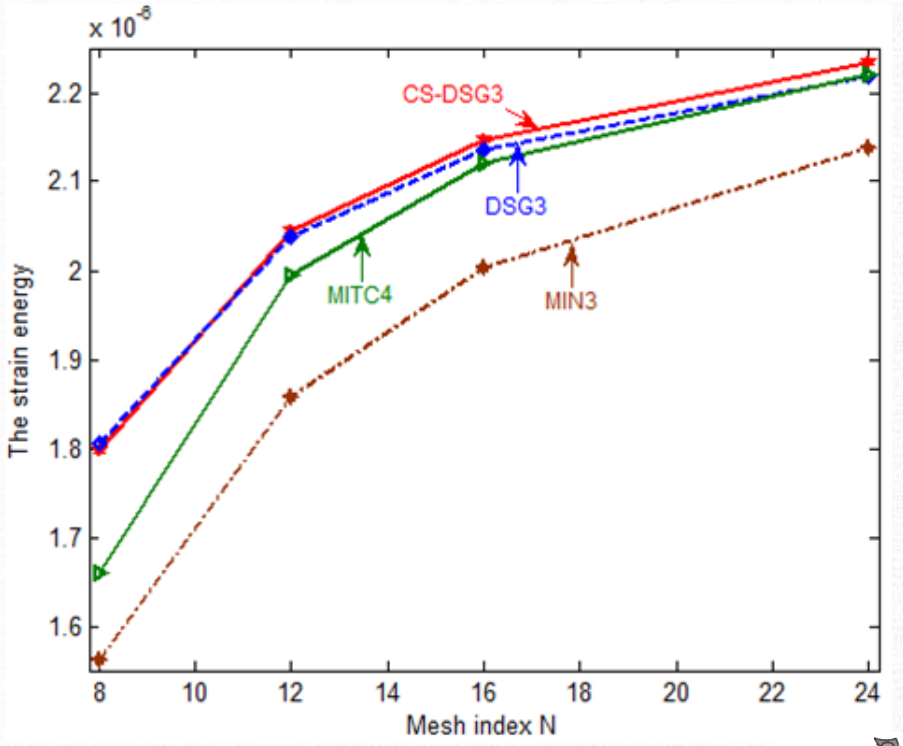




Plat shell

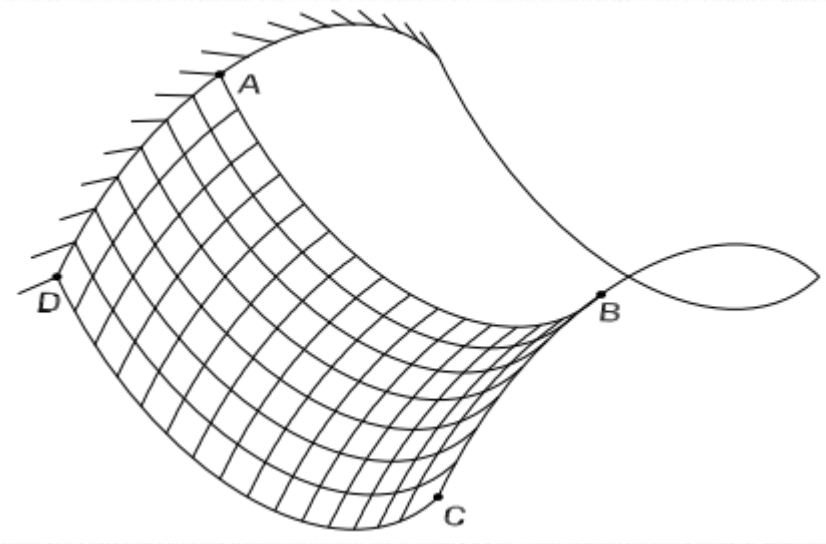
Pinched cylinder

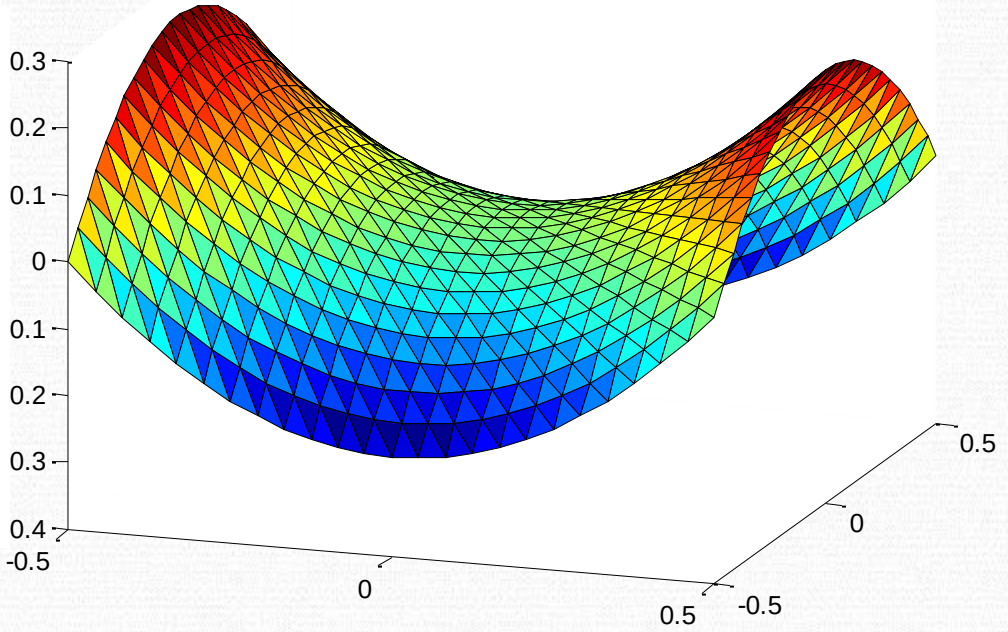




Plat shell

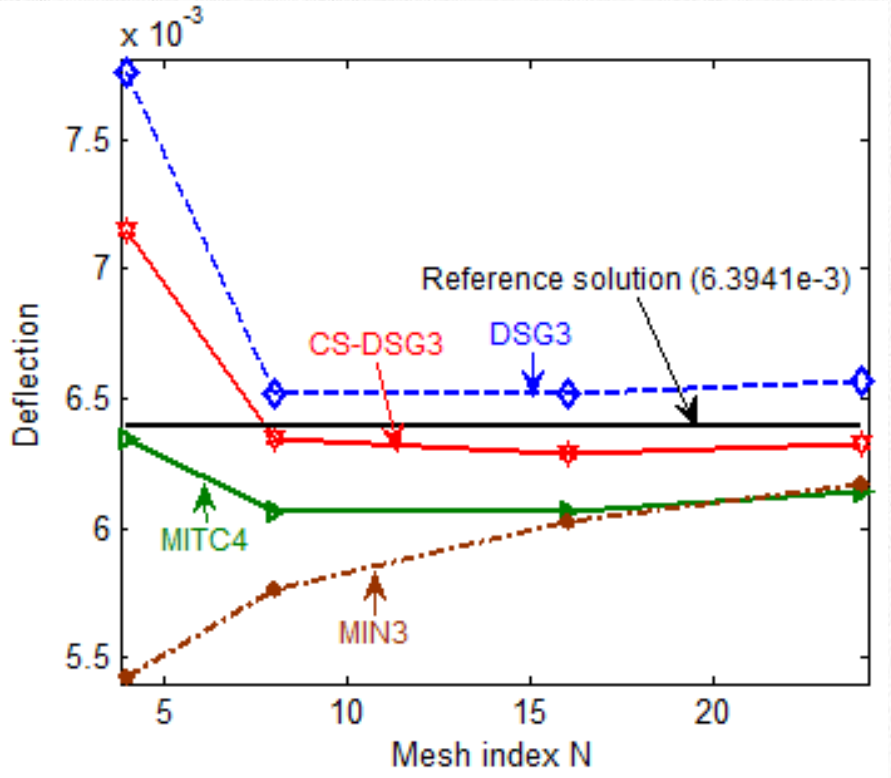
Party clamped hyperbolic paraboloid

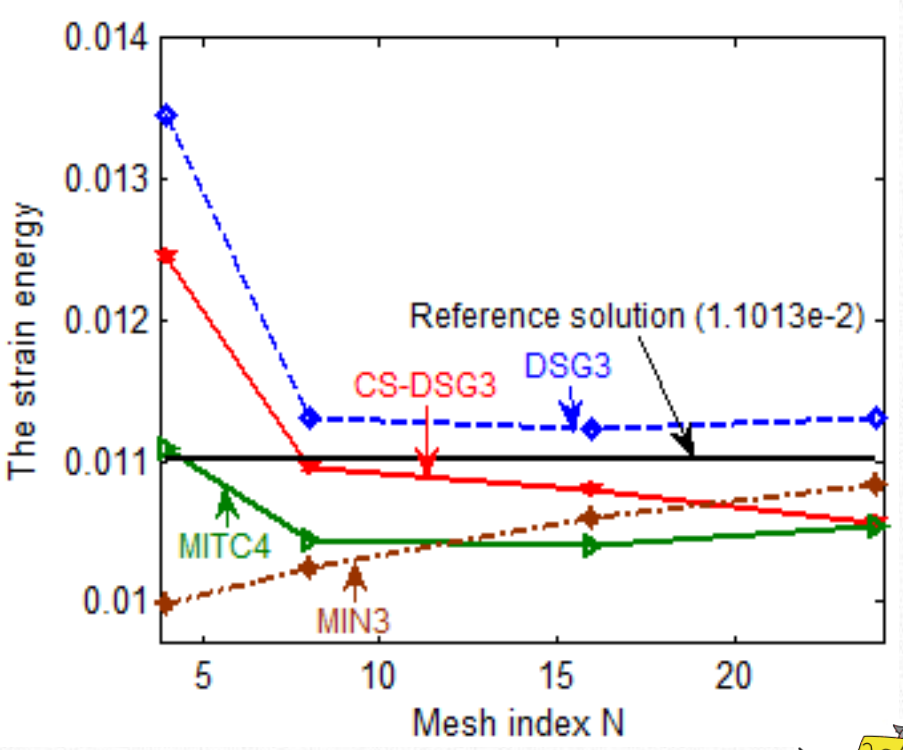




Plat shell

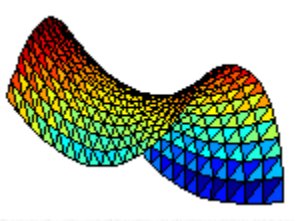
Party clamped hyperbolic paraboloid



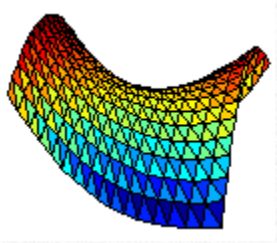


Plat shell

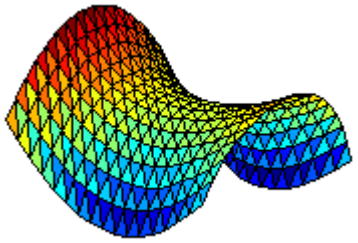
Party clamped hyperbolic paraboloid



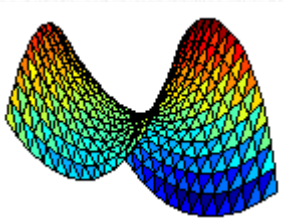
Mode 1



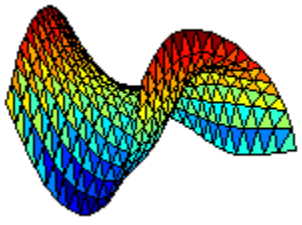
Mode 2



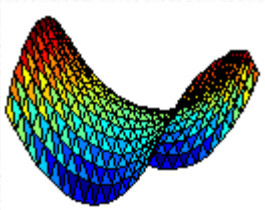
Mode 3



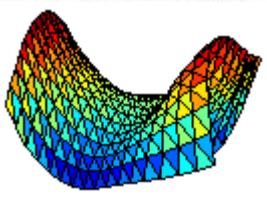
Mode 4



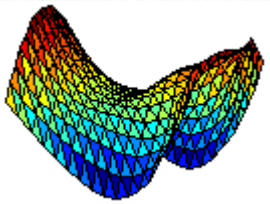
Mode 5



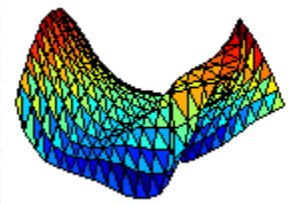
Mode 6



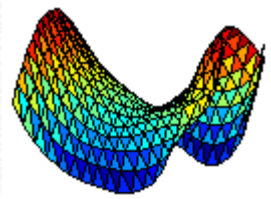
Mode 7



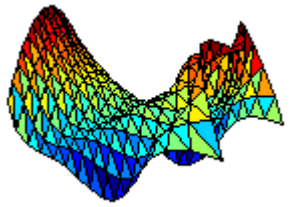
Mode 8



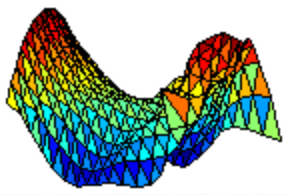
Mode 9



Mode 10



Mode 11



Mode 12

THANKS YOUR ATTENTION

